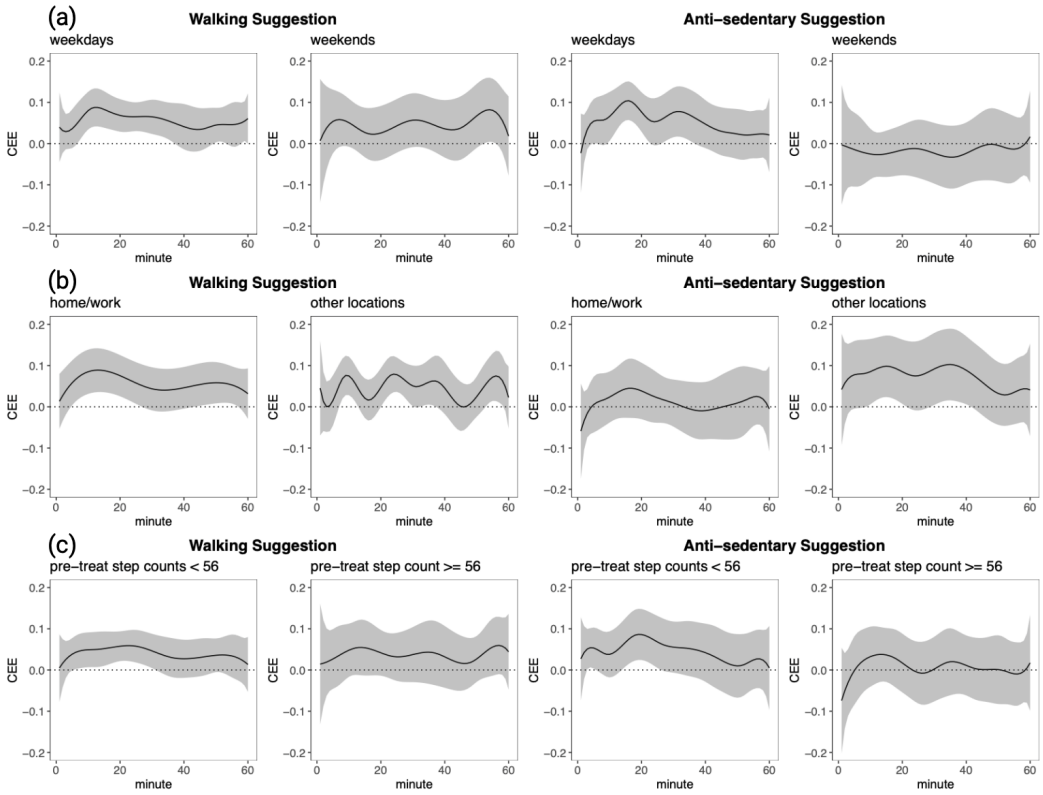

Figure 6: Estimated CEE of walking (left columns) and anti-sedentary (right columns) suggestions (compared to no suggestion) on minute-level step counts over a 60-minute period following the 35-th decision point (day 7), moderated by (a) location, (b) weekday/weekend, and (c) 30-min pre-treatment total step count.

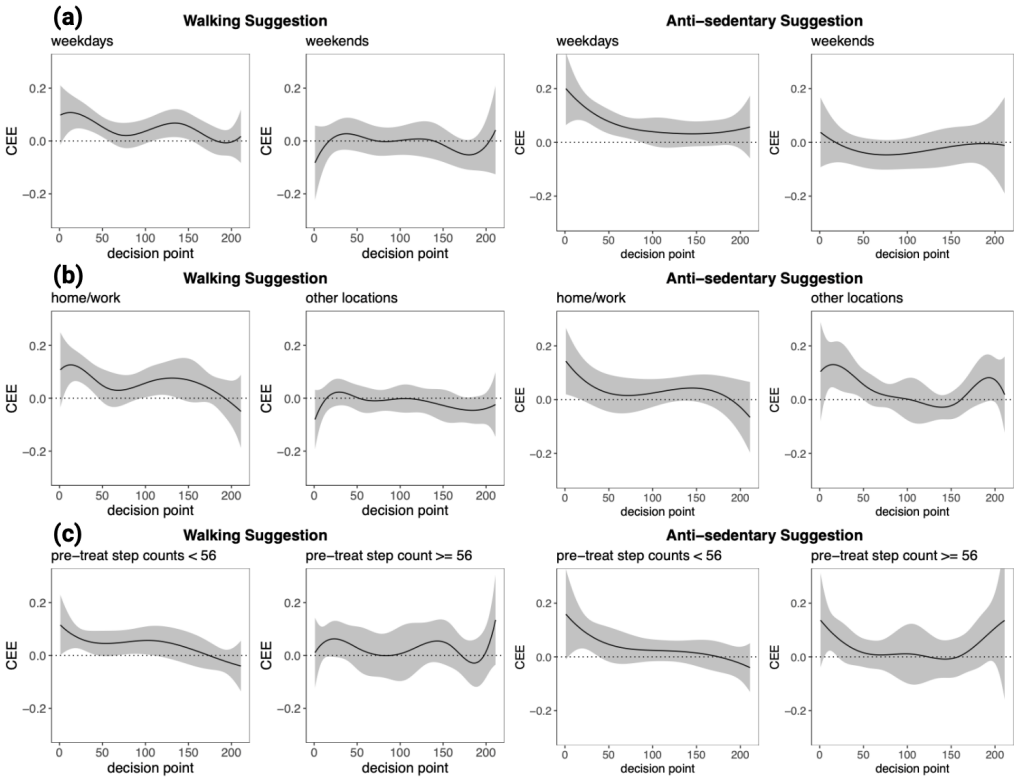

Figure 7: Estimated CEE of walking (left columns) and anti-sedentary (right rolumns) suggestions (compared to no suggestion) on the step count in the 15-th minute following each of the 210 decision point over the course of the study, moderated by (a) location, (b) weekday/weekend, and (c) 30-min pre-treatment total step count.

shows a significantly positive effect lasting for 40 minutes with a peak around minute 10-20 when $\text{sedentary}_{ij} = 1$, and this positive effect persisted through the first quarter of the study (see Figure 7(c)). This implies that it was easier to nudge the participants when they were in a sedentary state.

Figure 7 shows the CEE profile over decision points at minute 15 from these moderation analyses. Additional figures for other minutes are in the Supplementary Material, and they are vertical shifts of Figure 7. When the day was a weekday, when the participant was at home/work, or when the participant was recently sedentary, suggestions were effective in

promoting physical activity, especially during the first quarter of the study (up to around decision point 50). Comparing the anti-sedentary suggestion with the walking suggestion, the former started with a greater effect but had a faster decline over decision points, and thus its effect became insignificant earlier in the study. This indicates that participants may have become habituated to anti-sedentary suggestions particularly quickly, which may be alleviated in future studies by more variability in the suggestion content.

7.3 Causal Excursion Effect on Aggregate Outcome

Klasnja et al. (2018) found that walking suggestions had a positive effect on an aggregate proximal outcome, specifically the total step count within 30 minutes following a decision point. However, our double indices analyses in Section 7.1 suggest that this positive effect extends beyond the 30-minute window. This motivated us to replicate and extend the work of Klasnja et al. (2018), which focused on the 30-minute step count outcome. In this subsection, we explored aggregate outcomes over a longer time window to determine whether the 40-minute step count, as indicated by our analyses, shows a greater effect compared to the 30-minute total step count.

Specifically, we considered three aggregate outcomes: the total step count in a 30-minute, 40-minute, or 60-minute window following a decision point. For each of the outcomes, we applied the same analysis method used in Klasnja et al. (2018) and considered two estimands: (a) a fully marginal effect of the suggestion averaged over all decision points and all participants; and (b) a moderated effect with the decision point index being the effect modifier. We included the same control variables as the previous analyses for fitting a nuisance parameter in the weighted and centered least squares ($g_t(H_t)^T\alpha$ in (Boruvka et al., 2018)).